

NeuroFocus AI

A Multi-Modal Screening System for ADHD Analysis using Computer Vision and LLMs

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Abstract

This report presents NeuroFocus AI, a multi-modal behavioral screening system designed to identify potential indicators of ADHD. The system integrates computer vision, speech signal processing, and cognitive testing to extract objective behavioral markers. Unlike traditional approaches relying purely on subjective clinical observation, this system quantifies attention, cognitive load, and speech disfluencies using measurable signals, and combines them with LLM-based interpretation for meaningful insights.

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1 Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder characterized by inattention, impulsivity, and hyperactivity. Traditional diagnosis is time-intensive and subjective, often requiring clinical interviews and behavioral observation.

1.1 Problem Statement

There exists a gap between early symptom recognition and clinical diagnosis. Many individuals lack access to preliminary screening tools that can provide objective, data-driven insights into their behavioral patterns.

1.2 Proposed Solution

NeuroFocus AI addresses this gap by combining:

- Computer Vision (gaze tracking)
- Speech Signal Analysis (audio biomarkers)
- Cognitive Tasks (Stroop and memory tests)

The system focuses on **quantifiable behavioral markers** rather than subjective self-reporting.

2 System Architecture

The system follows a client-server architecture.

2.1 Frontend

Built using HTML, CSS, and JavaScript:

- Real-time face tracking via MediaPipe
- Cognitive task rendering
- Audio recording via browser APIs

2.2 Backend

Flask-based Python server:

- Audio preprocessing and feature extraction
- Speech-to-text conversion
- LLM integration (Gemini 2.5 Flash)
- SQLite database storage

3 Methodology

3.1 Speech Signal Processing

Unlike compressed formats such as MP3, WAV files store raw, uncompressed audio signals. This is critical because:

- Compressed formats remove subtle variations
- ADHD biomarkers depend on micro-variations in speech

Sound is a pressure wave. A microphone converts this into voltage signals, which are digitized via:

- **Sampling Rate:** Number of snapshots per second
- **Bit Depth:** Precision of each snapshot

3.1.1 Duration Calculation

Initially:

$$\text{Duration} = \frac{\text{Total Frames}}{\text{Frame Rate}}$$

Later improved using processed metadata:

$$\text{Duration} = \text{AudioSegment.duration.seconds}$$

3.1.2 RMS (Root Mean Square)

RMS represents perceived loudness:

$$RMS = \sqrt{\frac{1}{N} \sum x_i^2}$$

Normalized RMS:

$$RMS_{norm} = \frac{RMS}{32767}$$

Why RMS?

- Detects loudness variability
- Indicates emotional instability or cognitive stress

3.2 Speech Biomarkers

3.2.1 Words Per Minute (WPM)

Speech rate is calculated using transcription via SpeechRecognition.

Interpretation:

- High WPM → impulsivity
- Low WPM → cognitive hesitation

3.2.2 Silence Ratio

Defined as:

$$\text{Silence Ratio} = \frac{\text{Silence Duration}}{\text{Total Duration}}$$

Clinical Insight:

- < 10% → pressured speech (hyperactivity)
- > 30% → frequent pauses (inattention)

3.2.3 Disfluency Detection

Filler words like “uh”, “um” indicate cognitive load.

Why important? High filler frequency correlates with:

- Working memory overload
- Difficulty organizing thoughts

3.2.4 Feature Normalization

Weighted scoring:

- WPM: 40%
- Silence Ratio: 20%
- Filler Count: 20%
- RMS: 20%

Rationale: Speech rate is the strongest observable behavioral marker, hence highest weight.

3.3 Reaction Time Variability

$$\sigma = \sqrt{\frac{\sum (RT_i - \overline{RT})^2}{n - 1}}$$

Key Insight:

Two users may have same average, but:

- Consistent user → stable attention
- Variable user → attention lapses

ADHD is strongly associated with **high variability rather than mean performance**.

3.4 Vision-Based Attention Tracking

Using MediaPipe facial landmarks.

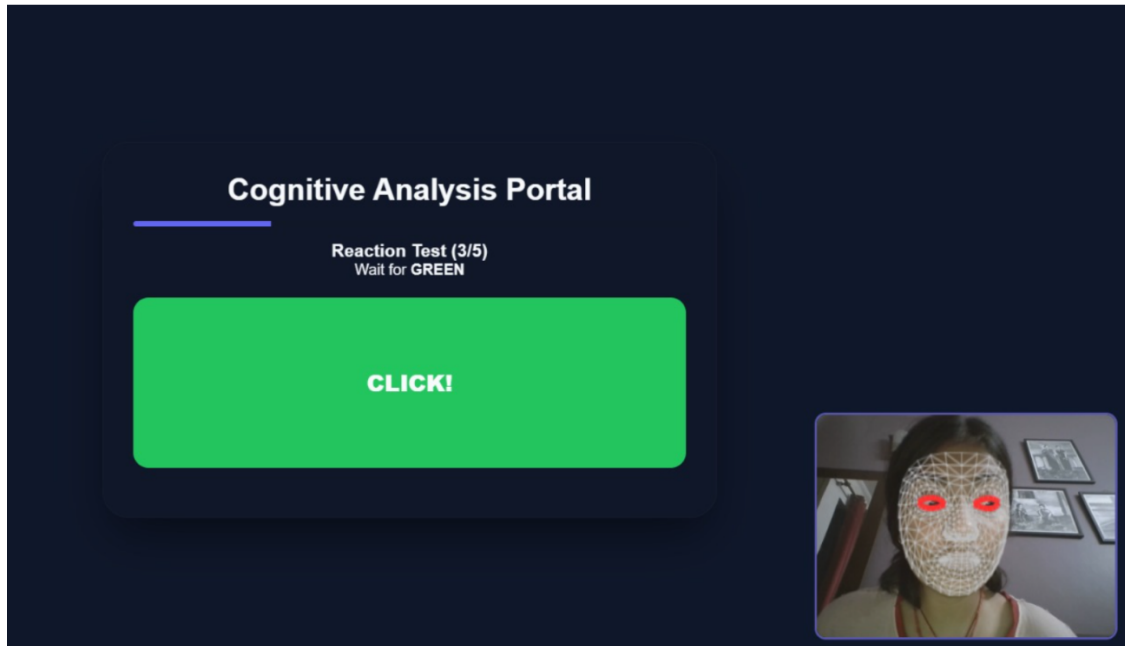


Figure 1: NeuroFocus AI Assessment Interface: Integrating the Reaction Time test with real-time MediaPipe face mesh tracking.

3.4.1 Eye Aspect Ratio (EAR)

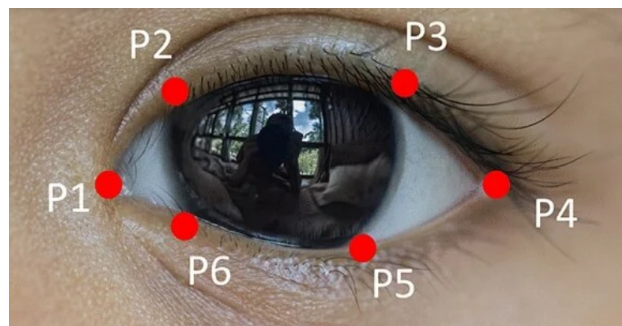


Figure 2: Eye landmark points used in EAR calculation. Vertical and horizontal distances determine eye state.

$$EAR = \frac{||p_2 - p_6|| + ||p_3 - p_5||}{2||p_1 - p_4||}$$

Interpretation:

- $EAR > 0.2 \rightarrow$ eyes open

- $\text{EAR} < 0.15 \rightarrow \text{eyes closed}$

Blink detection: Drop below threshold $\rightarrow$ rise again.

3.4.2 Gaze Distraction

$$(x > 0.65) \vee (x < 0.35) \vee (y < 0.35) \vee (y > 0.75)$$

Why this matters:

Attention lapses are not abstract—they manifest physically:

- Head turning
- Eye drifting
- Environmental distraction response

3.5 Cognitive Load Tasks

3.5.1 Stroop Test

Measures inhibitory control.

Observation: Users under cognitive load tend to:

- Look away
- Increase blink rate
- Physically fidget

3.5.2 Digit Span Test

Measures working memory capacity.

Low recall accuracy indicates:

- Poor short-term memory
- Reduced attention span

4 Design Decisions

4.1 Why WAV instead of MP3?

MP3 removes data $\rightarrow$ reduces accuracy of biomarkers.

4.2 Why Hardcoded Metrics before LLM?

LLMs can hallucinate. Extracting objective metrics ensures:

- Consistency
- Explainability

4.3 Why Multi-Modal Approach?

ADHD cannot be captured via a single signal. Combining:

- Vision
- Speech
- Cognitive tests

provides a more robust behavioral profile.

5 Implementation

The implementation of NeuroFocus AI involved a full-stack development approach, ensuring that high-frequency data from the frontend was processed efficiently by the Python backend.

5.1 Backend Orchestration (Flask)

The backend is built using the **Flask** framework. Its primary role is to act as a bridge between the user's raw behavioral data and the generative AI analysis.

- **Audio Processing:** Web browsers typically record audio in `.webm` format. The backend uses `pydub` to convert these files into `16-bit PCM WAV` files, which are essential for accurate RMS and WPM calculations.
- **Session Management:** Flask `sessions` are used to track the `assessment_id`, ensuring that cognitive scores and speech metrics are mapped to the correct user entry in the database.

5.2 LLM Integration (Google Gemini 2.5 Flash)

A core innovation of this project is the use of Large Language Models (LLMs) to provide qualitative clinical insights.

- **Model Selection:** Gemini 2.5 Flash was selected for its high-speed inference and "long context" capability, allowing it to "listen" to the 10-second audio clip and correlate it with the numerical metrics provided in the prompt.
- **Clinical Prompting:** The system uses a structured prompt that feeds the LLM objective data (Reaction Time Variability, Stroop Accuracy, and Silence Ratio). The model then generates a 3-sentence summary identifying whether the profile leans toward "Hyperactive" or "Inattentive" subtypes.

5.3 Data Persistence (SQLite)

To maintain data integrity, an **SQLite** database is used with a relational schema. This allows for historical tracking of assessments.

audio_id	assessment_id	duration	rms	gemin_report	wpm	transcript	silence_ratio	speech_risk_score	filter_count
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
11	11	13	4.68	0.0 As a clinical assistant, I am unable...	NULL	NULL	NULL	NULL	NULL
12	12	14	4.98 2889.07507324219	Processing...	NULL	NULL	NULL	NULL	NULL
13	13	15	4.98 2204.80123901367	Here's an analysis of the provided ...	NULL	NULL	NULL	NULL	NULL
14	14	16	4.86 2723.04244995117	**1. Tangentiality:** No evidence of...	NULL	NULL	NULL	NULL	NULL
15	15	24	4.74	0.0 1. **Correlation of Audio Recording...	0.0	[No speech detected]	1.0	0.8	0
16	16	25	4.74	0.0 **Correlation with Objective ...	0.0	[No speech detected]	1.0	0.8	0
17	17	27	5.76	0.0 Based on the objective metrics and ...	0.0	[No speech detected]	1.0	0.8	0
18	18	28	23.7	1407.138671875 Here's an analysis of the audio base...	0.0	[No speech detected]	0.899198312236287	0.8	0
19	19	29	6.54 4386.29556274414	1. **Correlation with Metrics:** Th...	110.091743119266	my name and when it was a normal day...	0.269113149847095	0.2	0
20	20	30	10.56 3157.53628540039	**Correlation with Metrics:**...	62.5	today every today everything went ...	0.466761363636364	0.8	0
21	21	31	9.18 3970.02813720703	Based on the objective metrics and ...	111.111111111111	so today everything it was a normal ...	0.372113289760349	0.4	0
22	22	32	9.72 4419.26684570313	**1. Correlation of Audio Recording ...	129.62962962963	so I woke up in the morning and a ...	0.0835390946502058	0.2	0
23	23	33	9.96	977.45703125 1. **Correlation of Audio with ...	0.0	[No speech detected]	0.754718875502008	0.8	0
24	24	34	9.96 1326.11828613281	The objective metrics accurately ...	0.0	[No speech detected]	0.854618473895582	0.8	0
25	25	35	9.96 1428.98858642578	**1. Correlation of Audio Recording ...	0.0	[No speech detected]	0.737248995983936	0.8	0
26	26	36	9.66 3362.48187255859	1. **Correlation of Audio with ...	49.6894409937888	it was a normal day everything went ...	0.561594202898551	0.8	0
27	27	37	9.96 839.259155273438	**1. Correlation with Audio ...	0.0	[No speech detected]	0.866566265060241	0.8	0
28	28	38	10.02 713.709106445313	The objective metrics accurately ...	0.0	[No speech detected]	0.953592814371258	0.8	0
29	29	39	9.96 706.67919921875	The objective metrics indicate a ...	0.0	[No speech detected]	0.953313253012048	0.8	0
30	30	40	9.96 706.594390869141	The provided metrics strongly ...	0.0	[No speech detected]	0.953815261044177	0.8	0
31	31	41	5.88 3289.48095703125	**1. Correlation of Audio Recording ...	81.6326530612245	it was a normal day everything went ...	0.562244897959184	0.4	0
32	32	43	12.84 4362.87518310547	Here's an analysis of the provided ...	60.7476635514019	hello hello can you record my voice ...	0.45911214953271	0.8	0
33	33	44	4.92 4028.16732788086	The slow speech rate of 85.4 WPM and...	85.3658536585366	hello can you listen to my voice	0.24369918699187	0.2	0

Figure 3: Relational database view of the **speech_analysis** table, showing the the metrics recorded and calculated risk levels for various assessment sessions.

assessment_id	full_summary	risk_level	generated_at	score
Filter	Filter	Filter	Filter	Filter
1	30 The high reaction time variability, ...	NULL	2026-03-31 08:25:07	0
2	31 The elevated Reaction Time ...	NULL	2026-03-31 08:49:01	0
3	35 The cognitive scores demonstrate ...	NULL	2026-03-31 08:58:25	0
4	39 The extreme lack of verbal output, ...	NULL	2026-03-31 09:01:11	0
5	41 The cognitive data, showing high ...	NULL	2026-03-31 09:08:37	0
6	44 **Clinical Summary:**...	Moderate	2026-04-02 13:36:16	2

Figure 4: Relational database view of the **final_reports** table, showing the generated clinical summaries and calculated risk levels for various assessment sessions.

6 Future Scope

- Use variance of speech features (burstiness detection)
- Chunk audio into 1-second windows for temporal analysis

- Mobile optimization
- Larger dataset training

7 Conclusion

NeuroFocus AI demonstrates that ADHD screening can be transformed into a quantifiable, data-driven process. By combining signal processing, behavioral science, and AI, the system bridges the gap between subjective observation and objective measurement.

Disclaimer

This tool is for screening purposes only and does not replace professional medical diagnosis.